

ML Graded Quiz M6

Question 1

When will a linear model tend to outperform a neural network?

- ☐ Linear models are generally complex algorithms and would therefore discover patterns in our dataset.
- ☐ If the data size is small
- ☐ Neural networks are still far from being used in the industry.
- ☒ If the data size is large

Question 2

What is the most important characteristic of a good credit scorecard model?

- ☐ An explainable model
- ☐ A model with better precision
- ☐ A model with better accuracy
- ☒ A model with high AUROC

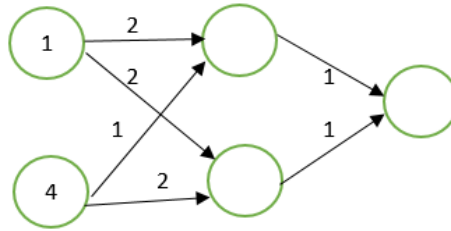
Question 3

Which is **not** a challenge of optimization in deep learning?

- ☐ Local minima
- ☐ Saddle points
- ☐ Exploding gradient
- ☒ Vanishing gradient

Question 4

Calculate the error (predicted - target) we get if we predict the output with the given weights and the input if the actual target value is 3.



- ☐ 7
- ☐ 0
- ☐ 13
- ☒ 16

Question 5

What is the purpose of the scoring variable when performing cross-validation?

- ☐ It is the backpropagation holder.
- ☐ It is the loss function.
- ☐ It is the batch size.
- ☒ It is the activation function.

Question 6

Why do we split our data into training and validation sets?

- ☐ Because it allows a maximum performance score on the training set.
- ☐ Because it speeds up model training.
- ☐ Because it prevents any data leakage from the test set to the training set and therefore provides a better generalization of the model.
- ☒ Because it prevents underfitting of the model and therefore ensures the model is robust.

Question 7

At all points of the real line, which of the functions below is **not** a convex function?

- ☐ $-x$
- ☐ $5x^4 - 9x^2$
- ☐ e^{-x}
- ☒ $5x^2 - 4x + 3$

Question 8

Find the dimensions of the inputs (x), the weights (W), and the bias (b) such that the output takes the form ($Wx + b$) in the table below.

ID	x_1	x_2	x_3	$\dots$	x_n	y
1	7	6	5		8	1
2	1	0	2		1	0
3	8	7	9		3	1
$\vdots$						
$n-1$	2	3	1		4	0
n	3	4	3		3	0

- ☐ $W : (1 \times n), x : (n \times 1), b : (1 \times 1)$
- ☐ $W : (n \times 1), x : (1 \times n), b : (1 \times 1)$
- ☐ $W : (n \times 1), x : (1 \times n), b : (n \times 1)$
- ☒ $W : (1 \times n), x : (1 \times n), b : (n \times 1)$

Question 9

When forecasting yield curves using a multi-linear perceptron, what is the most suitable **metric** to measure the performance of the model?

- ☐ Classification metrics
- ☐ Accuracy
- ☐ Categorical entropy
- ☒ Root mean squared error

Question 10

Assume a borrower takes a mortgage for \$475,000 and the bank finances **90%** of the purchase. If the borrower manages to repay \$72,000, what is the exposure at default if the historical default rate is **5%**?

- ☐ \$427,500
- ☐ \$17,750
- ☐ \$21,375
- ☒ \$355,500

Question 11

Which of the following is **correct**?

- ☐ When a function is not differentiable, we can't say whether it is convex or not.
- ☐ The second derivative of a function might tell us whether a function is convex or not.
- ☐ The gradient of a function can tell us whether a function is convex or not.
- ☒ All polynomial functions of degree 2 are convex functions.

Question 12

Find the slope vector if the input data is **(2, 2, 3)**, the weight is **(0, 2, 1)** and the target is 4. Assume the loss function is a mean squared error $J = \sum_{i=1}^n (y_i - \hat{y})^2$.

- ☐ (28, 56, 42)
- ☐ (12, 12, 18)
- ☐ (16, 16, 24)
- ☒ (28, 28, 42)

Question 13

Which is the least likely to be an application of yield curves?



Time left:
1:59:38

QUESTIONS

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8

- ☐ Indicating trade-offs between maturity and yield
- ☐ Determining the volatility of options
- ☐ Detecting overpriced and underpriced securities
- ☒ Forecasting interest rates

9	10	11	12
13	14	15	16

Question 14

The decision to grant an additional loan to an existing client can be modeled by which type of model?

- ☐ Behavioral model
- ☐ Application model
- ☐ Survival model
- ☒ Collection model

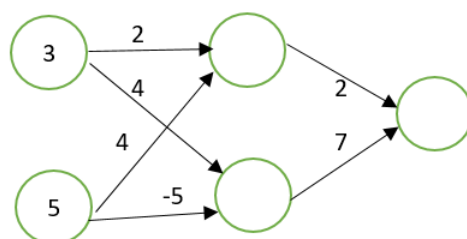
Question 15

Which of these is **not** a convex set?

- ☐ Torus
- ☐ Cube
- ☐ Tennis ball
- ☒ Triangle

Question 16

Calculate the output of the neural network shown below. Apply the ReLU activation function at the hidden layer and the identity activation function at the output layer.



- ☐ 39
- ☐ -39

☒ 52

☐ 0

SUBMIT

